

we're going to start now



Day 1: Bren Math Workshop

Carmen Galaz García, Ph.D.

Bren School of Environmental Science & Management

MESM Math Workshop basics

Welcome to the MESM Math Workshop

- **Instructor:** Carmen Galaz García
- **Course hours:** Daily 12:00pm - 2:00pm PST (except for Day 5, which is 12:30pm - 2:30pm PST)
- **Location:** Bren Hall 1414
- **Website:** <https://bit.ly/mesm-math-workshop>



About your instructor

Carmen Galaz García (she/her)

- Assistant Teaching Professor @ Bren

Before:

- Data Scientist @ NCEAS
- Ph.D. in Mathematics @ UCSB



Teaching & Research

- Python courses and Capstone Projects for Masters in Environmental Data Science (MEDS)
- Supporting math and data science across Bren!
- NCEAS/TNC collaboration for invasive iceplant mapping analyzing remotely sensed images



About you!



OK!!!
let's
wrap it
up!



Topics overview

- **Day 1:** Algebra refresher; Rules of exponents
- **Day 2:** Units; Functions; Graphing;
- **Day 3:** Exponential functions; Logarithms; Linear functions
- **Day 4:** Limits & Derivatives
- **Day 5:** Integration & ODEs

Workshop Objectives

1. 🧹 Shake off the math dust
2. 💪 Exercise core math skills needed to succeed in your Bren courses
3. 🌍 Explore how valuable math is to environmental data science
4. 🤝 Build a collaborative environment and get to know your cohort

Student expectations

We expect all students will:

- 🤗 Support and encourage all classmates
- 🌱 Be open to learning from each other
- 🙌 Bring a collaborative attitude and communicate respectfully
- 💡 See opportunities to share knowledge as a chance to deepen understanding
- ✅ Complete all in-class exercises

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Logistics

Please fill out this survey



Or you can click on this link.

Math in Environmental Science

Why am I taking a math class?

or,

*Why do I have to know math when a
computer will do it for me?*

A quantitative brain warm-up

MESM students have diverse work & academic histories

This course (re)introduces math concepts and tools that:

- Focus on applications to environmental data science
- Are specifically relevant for MEDS projects, coursework
- Provide an entryway into building computational skills
- Refresh quantitative thinking skills generally
- Refresh essentials like units, conversions, notation, **language**

Share with the person next to you:

What has been your life or professional experience with math? Best friend ever, mortal enemy, or something in between?

How do you see math being relevant to solving environmental problems?

Math is an important tool in environmental science

Math helps us investigate the world and communicate our findings

- Used by scientists to find evidence
- Can be used to quantify relations, trends, and changes
- Mathematical models can help us understand complex systems
- Must be used responsibly and with awareness of potential biases and limitations
- Helps us support arguments that can transform policy and actions

We need to understand the math to responsibly make decisions on it!

Math at Bren

In classes:

- ESM 201: Ecology of Managed Ecosystems

Lotka-Volterra Models

$$\begin{aligned}\frac{dN_1}{dt} &= r_1 N_1 \left(\frac{K_1 - N_1 - \alpha N_2}{K_1} \right) \\ \frac{dN_2}{dt} &= r_2 N_2 \left(\frac{K_2 - N_2 - \beta N_1}{K_2} \right)\end{aligned}$$

- ESM 222: Pollution Risk Management

Groundwater transport of absorbed contaminant

$$\frac{\partial C}{\partial t} = \left(\frac{D}{R} \frac{\partial^2 t}{\partial x^2} \right) - \left(\frac{v}{R} \frac{\partial C}{\partial x} \right) - \frac{k}{R} C$$

In Research

- Ecology, modeling, economics, and more!
- Applying Portfolio Theory (A mix of calculus and stochastic linear algebra) to salmon stocks (MESM GP)
- Quantifying Greenhouse Gas (GHG) Emissions Associated with Global Seafood Production (MEDS Capstone)

Today: Math brain warm up

- Algebra review
- Rules of exponents
- Applications

Algebra review

Rules of algebra

You can get far with a few rules:

1. Follow order of operations (a.k.a. PEDMAS)
2. Never change an equation. Instead, rewrite them into more useful forms
3. Manipulate BOTH sides of an equation with the SAME PROCEDURE.

Always add/subtract/multiply/divide by the same number on both sides. More generally, always apply the same function to both sides of the equation.

Order of operations (P-E-DM-AS)

P - Parentheses

E - Exponents

D/M - Division / multiplication

A/S - Addition / subtraction

Order of operations practice problems

 Take a minute to solve these individually.

- Simplify $\frac{4-6}{2}(3+1) - \frac{1+2*4}{3}$
- Simplify $3x + 4(8x - 6x) - (2y - 5) + \frac{2x(1-3)}{2}$
- If we are given the values of $x = -3$ and $y = 2$, what would be the value of z ?

$$z = 4 * (y - 4) + (x + 1)^2$$

Order of operations practice problems

💡 Let's see a solution!

$$\frac{4-6}{2}(3+1) - \frac{1-2 \cdot 4}{3} = -7$$

$$\frac{-2}{2} 4 - \frac{1+8}{3}$$

$$= -4 - \frac{9}{3}$$

$$= -4 - 3$$

$$= \underline{-7.}$$

$$3x + 4(8x - 6x) - (2y - 5) + \frac{2x(1-3)}{2}$$

$$\underline{3x + 32x - 24x - 2y + 5} + \underline{x - 3x}$$

$$\underline{35x - 24x - 2y + 5 - 2x}$$

$$\underline{11x - 2y + 5 - 2x}$$

$$9x - 2y + 5$$

Order of operations practice problems

💡 Let's see a solution!

If we are given the values of $x = -3$ and $y = 2$, what would be the value of z ?

$$z = 4 * (y - 4) + (x + 1)^2$$

$$z = 4(2 - 4) + (-3 + 1)^2$$

$$= 4(-2) + (-2)^2$$

$$= -8 + 4$$

$$= \underline{\underline{-4}}.$$

Practice solutions

$$\begin{aligned}\frac{4-6}{2}(3+1) - \frac{1+2*4}{3} &= \frac{-2}{2}(4) - \frac{9}{3} \\ &= -1(4) - 3 \\ &= -4 - 3 \\ &= -7\end{aligned}$$

$$\begin{aligned}3x + 4(8x - 6x) - (2y - 5) + \frac{2x(1-3)}{2} &= 3x + 4(2x) - 2y + 5 + \frac{2x - 6x}{2} \\ &= 3x + 8x - 2y + 5 + \frac{-4x}{2} \\ &= 3x + 8x - 2y + 5 - 2x \\ &= 9x - 2y + 5\end{aligned}$$

$$\begin{aligned}z &= 4(y-4) + (x+1)^2 = 4(2-4) + (-3+1)^2 \\ &= 4(-2) + (-2)^2 \\ &= -8 + 4 \\ z &= -4\end{aligned}$$

Solving equations

Equations help us find unknown values that satisfy certain conditions.

Remember the rules:

2. Never change an equation. Instead, we rewrite them into more useful forms
3. Manipulate BOTH sides of an equation with the SAME PROCEDURE.

Always add/subtract/multiply/divide by the same number on both sides. More generally, always apply the same function to both sides of the equation.

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Solving equations: example

 Isolate y in terms of x :

$$x = \frac{(400 - y)}{80}$$

$$80x = \cancel{80} \frac{(400 - y)}{\cancel{80}}$$

$$80x = 400 - y$$

$$80x + y = 400 - \cancel{y} + \cancel{y}$$

$$80x + y = 400$$

$$\cancel{-80x} + \cancel{80x} + y = 400 - 80x$$

$$\underline{y = 400 - 80x}$$

Solving equations: example

💡 Let's see a solution!

Isolate y in terms of x :

$$x = \frac{(400 - y)}{80}$$

$$80x = \frac{(400 - y)\cancel{80}}{\cancel{80}}$$

Multiply both sides by 80

$$80x - 400 = \cancel{400} - \cancel{400} - y$$

Subtract both sides by 400

$$-1(80x - 400) = -y(-1)$$

Multiply both sides by -1

$$400 - 80x = y$$

Flip terms for simplicity

Solving equations: Practice set 1

It can be easy to make mistakes while doing algebra. Practice is key!

 Solve all in terms of x :

a. $3x + 2 = 10x - 12$

b. $4 - 3(2x + 1) = 8 - \frac{3x}{2}$

c. $3(x + 7a) - 5 = b + 2(c - 4x)$

Solving equations: Practice set 1

 Let's see a solution!

Solve all in terms of x

$$\text{a. } 3x + 2 = 10x - 12$$

$+12$ $+12$

$$\begin{array}{r} 3x + 14 = 10x \\ -3x \quad -3x \end{array}$$

$$\frac{14}{7} = \frac{\cancel{7}x}{\cancel{7}}$$

$$\boxed{x = 2}$$

$$\text{b. } (4 - 3(2x + 1) = 8 - \frac{3x}{2}) \cdot 2$$

$$8 - 6(2x + 1) = 16 - 3x$$

$$\begin{array}{r} 8 - 12x - 6 = 16 - 3x \\ -16 \quad +12x \quad -16 \quad +12x \end{array}$$

$$9x = 2 - 16$$

$$9x = -14$$

$$x = -\frac{14}{9}$$

Solving equations: Practice set 1

 Let's see a solution!

Solve all in terms of x

c. $3(x + 7a) - 5 = b + 2(c - 4x)$

$$3x + 21a - 5 = b + 2c - 8x$$

$$3x = -21a + 5 + b + 2c - 8x$$

$$11x = -21a + b + 2c + 5$$

$$x = \frac{-21a + b + 2c + 5}{11}$$

Practice set 1 solutions

$$\begin{aligned}3x + 2 &= 10x - 12 \\3x + 2 + 12 &= 10x - 12(+12) \\3x - 3x + 14 &= 10x - 3x \\14 &= 7x \\x &= 2\end{aligned}$$

$$\begin{aligned}4 - 3(2x + 1) &= 8 - \frac{3x}{2} \\4 - 3 - 6x &= 8 - \frac{3x}{2} \\1 - 6x &= 8 - \frac{3x}{2} \\2 - 12x &= 16 - 3x \\-9x &= 14 \\x &= \frac{-14}{9}\end{aligned}$$

Practice set 1 solutions

$$3(x + 7a) - 5 = b + 2(c - 4x)$$

$$3x + 21a - 5 = b + 2c - 8x$$

$$11x + 21a - 5 = b + 2c$$

$$11x = 5 + b + 2c - 21a$$

$$x = \frac{5 + b + 2c - 21a}{11}$$

Solving equations: Practice set 2



1. Prices are important in economics, but not always available for environmental goods. If we know quantity $Q = \frac{(400 - P)}{80}$, isolate P in terms of Q .

2. The logistic per-capita growth rate is $r \left(1 - \frac{N}{K} \right)$, where N is population size, K is carrying capacity, and r is the intrinsic growth rate. If this rate equals 0.02 when $N = 750$ and $K = 1000$, solve for r .

3. A watershed's water balance equation is:

$$\text{change in storage per unit time} = \text{precipitation } (P) - \text{actual evapotranspiration (ET)} - \text{river outflows } (Q).$$

At steady state (storage isn't changing), actual evapotranspiration (ET) is 60% of precipitation and river outflow is 320 mm/y. Solve for precipitation.

Solving equations: Practice set 2

💡 Let's see a solution!

1) Prices are important in economics, but not always available for environmental goods. If we know quantity $Q = \frac{(400 - P)}{80}$, isolate P in terms of Q .

$$Q = \frac{(400 - P)}{80} \cdot 80$$

$$\begin{array}{rcl} 80Q & = & 400 - P \\ -400 & & -400 \end{array}$$

$$(-1)(80Q - 400) = -P \quad (-1) \quad \rightarrow \quad -80Q + 400 = P$$

Solving equations: Practice set 2

💡 Let's see a solution!

2) The logistic per-capita growth rate is $r \left(1 - \frac{N}{K} \right)$, where N is population size, K is carrying capacity, and r is the intrinsic growth rate. If this rate equals 0.02 when $N = 750$ and $K = 1000$, solve for r .

$$r \left(1 - \frac{750}{1000} \right) = 0.02$$

$$r (1 - 0.75) = 0.02$$

$$r (0.25) = 0.02$$

$$0.25r = 0.02$$

$$\frac{0.25r}{0.25} = \frac{0.02}{0.25}$$

$$r = 0.08$$

Solving equations: Practice set 2

💡 Let's see a solution!

3) A watershed's water balance is: change in storage per unit time = precipitation (P) – actual ET – river outflows (Q). At steady state (storage isn't changing), with actual evapotranspiration (ET) being 60% of precipitation and $Q = 320$ mm/y, solve for P .

$$0 = P - 0.6P - Q$$

$$\begin{array}{rcl} 0 & = & 0.4P - Q \\ +Q & & +Q \end{array}$$

$$0.4P = Q$$

$$\frac{0.4P}{0.4} = \frac{320 \text{ mm}}{0.4 \text{ mm/y}} = P = 800 \text{ mm/y}$$

Practice set 2 solutions

1)

$$Q = \frac{(400 - P)}{80}$$

$$80Q = 400 - P$$

$$80Q - 400 = -P$$

$$P = 400 - 80Q$$

2)

$$1 - \frac{750}{1000} = 0.25$$

$$0.25r = 0.02$$

$$r = 0.08 \text{ per year}$$

3)

$$\text{storage} = P - AET - Q$$

$$0 = P - 0.6P - Q$$

$$0.4P = 320$$

$$P = 800 \text{ mm/y}$$

Let's take a 10 minute break



image: Flaticon.com

Rules of exponents

Exponents make algebra more interesting!

Example

The expression x^3 means x multiplied by itself 3 times:

$$\underline{x^3} = \underline{x} * \underline{x} * \underline{x}$$

More generally,

$$\underline{x^n} = \underline{x \text{ to the power of } n.}$$

Intuitively, the **exponent** n tells us how many times to multiply the **basis** x by itself:

$$x^n = \underbrace{x \cdot x \cdot x \cdot \dots \cdot x}_{n \text{ times}} \quad x^n = \underbrace{x \cdot x \cdot x \cdot \dots \cdot x}_{n \text{ times.}}$$

But the exponent can be any (real) number, not only integers!

What are exponents good for?

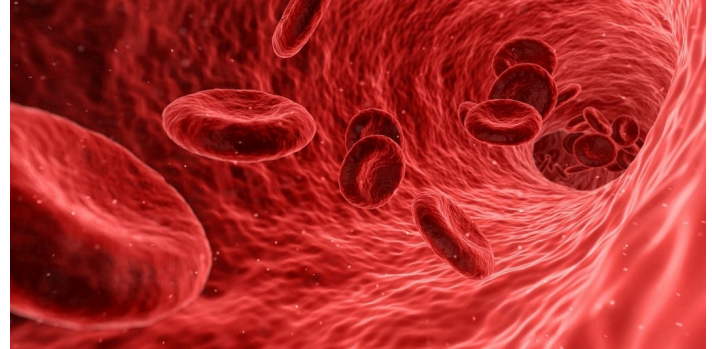
They can describe really big numbers



Diameter of the Milky Way =
1,000,000,000,000,000,000,000 meters =

$$10^{21} \text{ m}$$

And also really small numbers



Diameter of a human red blood cell =
0.0000000001 meters =

$$10^{-10} \text{ m}$$

This way, we can use exponents to describe exponential growth and decay. Many environmental variables behave in this way.

Rules and properties of exponents (1)

1. Product rule: $x^a \cdot x^b = x^{a+b}$.



2. Quotient rule: $\frac{x^a}{x^b} = x^{a-b}, x \neq 0$.

* 3. Any non-negative x to the power of 0 equals 1: $x^0 = 1$, for $x \neq 0$

4. Negative exponent means divide: $x^{-a} = \frac{1}{x^a}$

5. Fractional exponent means take the n -th root:

$$x^{\frac{1}{n}} = \sqrt[n]{x}.$$

Rules and properties of exponents (2)

6. Power of a power means multiply the exponents: $(x^a)^b = x^{a \cdot b}$



7. Exponents get distributed over products: $(xy)^a = x^a \cdot y^a$

8. And exponents get distributed over divisions: $\left(\frac{x}{y}\right)^a = \frac{x^a}{y^a}$

9. If a is the n -th root of x , then a to the n equals x :

$$a = \sqrt[n]{x} \Rightarrow a^n = x.$$

Rules and properties of exponents (1)

1. Product rule: $x^a * x^b = x^{a+b}$

2. Quotient rule: $\frac{x^a}{x^b} = x^{a-b}$, for $x \neq 0$

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Rules and properties of exponents (2)

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7. Exponents get distributed over products: $(xy)^a = x^a y^a$

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9. If a is the n -th root of x , then a to the n equals x :

$$a = \sqrt[n]{x} \Rightarrow a^n = x$$

Rules and properties of exponents

Rule	Expression
Product Rule	$x^a \cdot x^b = x^{a+b}$
Quotient Rule	$\frac{x^a}{x^b} = x^{a-b}, \quad x \neq 0$
Zero Exponent	$x^0 = 1, \quad x \neq 0$
Negative Exponent	$x^{-a} = \frac{1}{x^a}, \quad x \neq 0$
Fractional Exponent	$x^{\frac{1}{n}} = \sqrt[n]{x}$
Power of a Power	$(x^a)^b = x^{ab}$
Power of a Product	$(xy)^a = x^a y^a$
Power of a Quotient	$\left(\frac{x}{y}\right)^a = \frac{x^a}{y^a}$
n -th Root Definition	$a = \sqrt[n]{x} \Rightarrow a^n = x$

Exponents practice

 Simplify the following.

a. $\frac{6^5}{6^3} = 6^{5-3} = 6^2 = 36.$

b. $(x^2y)^4 = (x^2)^4 \cdot y^4 = x^8 \cdot y^4.$

c. $\frac{x^5y^6}{x^2y^2} = \frac{x^5}{x^2} \cdot \frac{y^6}{y^2} = x^{5-2} \cdot y^{6-2} = x^3 \cdot y^4 = (xy)^4.$

d. $\frac{24x^6}{12x^{-8}} = \frac{24}{12} \cdot \frac{x^6}{x^{-8}} = 2 \cdot x^{6-(-8)} = 2 \cdot x^{6+8} = 2x^{14}.$

Rule	Expression
Product Rule	$x^a \cdot x^b = x^{a+b}$
Quotient Rule	$\frac{x^a}{x^b} = x^{a-b}, \quad x \neq 0$
Zero Exponent	$x^0 = 1, \quad x \neq 0$
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Fractional Exponent	$x^{\frac{1}{n}} = \sqrt[n]{x}$
Power of a Power	$(x^a)^b = x^{ab}$
Power of a Product	$(xy)^a = x^a y^a$
Power of a Quotient	$\left(\frac{x}{y}\right)^a = \frac{x^a}{y^a}$
n -th Root Definition	$a = \sqrt[n]{x} \Rightarrow a^n = x$

Exponents practice

 Let's see a solution.

a. $\frac{6^5}{6^3} = 6^{5-3} = 6^2$

b. $(x^2y)^4 = (x^2)^4 * y^4 = x^8y^4$

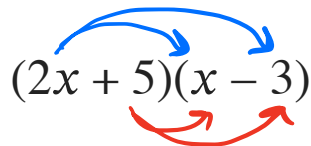
c. $\frac{x^5y^6}{xy^2} = \frac{x^5}{x} * \frac{y^6}{y^2} = x^{5-1} * y^{6-2} = x^4y^4 = (xy)^4$

d. $\frac{24x^6}{12x^{-8}} = 2x^6x^{(-(-8))} = 2x^6x^8 = 2x^{6+8} = 2x^{14}$

Multiplying expressions (FOIL)

First, **O**utside, **I**nside, **L**ast

Example:

$$(2x + 5)(x - 3)$$


 Expand this expression.

$$\begin{aligned} & 2x \cdot x + 2x(-3) + 5x + 5(-3) \\ &= 2x^2 - 6x + 5x - 15 \\ &= \underline{2x^2 - x - 15.} \end{aligned}$$

Multiplying expressions (FOIL)

First, **O**utside, **I**nside, **L**ast

Example:

$$(2x + 5)(x - 3)$$

 Let's see a solution.

$$= (2x \times x)(2x \times -3)(5 \times x)(5 \times -3)$$

$$= 2x^2 - 6x + 5x - 15$$

Polynomials describe more complex relationships through exponents

A polynomial is a function of the form:

$$a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x^1 + a_0$$

- x is the **variable**
- a_i are **constants** (fixed numbers) that are called the **coefficients** of the polynomial
- each $a_i x^i$ is called a **term** of the polynomial,
- n is called the **degree** of the polynomial, this is the largest exponent in the polynomial.

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- each $a_i x^i$ is called a **term** of the polynomial,
- n is called the **degree** of the polynomial, this is the largest exponent in the polynomial.

Example:

- $x^5 + x^2 - 1$ is a polynomial of degree five with three terms.
- Linear functions like $mx^1 + c$ where m and c are constants are polynomials of degree 1!

Roots of polynomials

✎ A zero, root, solution for a polynomial is a value of x that makes the polynomial equal to 0.

A polynomial of degree n , it will have exactly n roots.

Roots of polynomials

A **zero**, **root**, or **solution** for a polynomial is a value of x that makes the polynomial equal to zero.

Fundamental Theorem of Algebra

A polynomial

$$a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$$

of degree n has exactly n roots $r_1, \dots, r_n$ and it can be refactored as

$$c(x - r_1) \dots (x - r_n).$$

The numbers c and $r_1, \dots, r_n$ may be complex!

Example:

Consider the function $x - 2$, this is just a 1st degree polynomial. It has a single root, $r_1 = 2$.

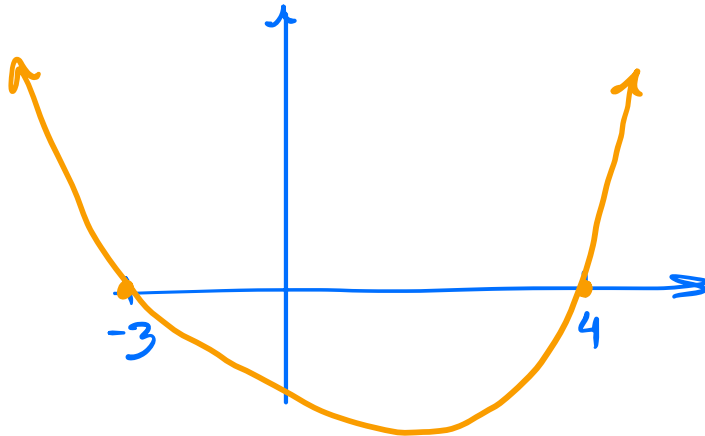
Example: second degree polynomial

✎ Check: What happens if $x = 4$ or $x = -3$ in $x^2 - x - 12$?

Check that $x=4$ and $x=-3$ are roots:

$$4^2 - 4 - 12 = 16 - 4 - 12 = 12 - 12 = 0 \Rightarrow 4 \text{ is a root}$$

$$(-3)^2 - (-3) - 12 = 9 + 3 - 12 = 12 - 12 = 0 \Rightarrow -3 \text{ is a root.}$$



$(4, 0)$

$(-3, 0)$

Example: second degree polynomial

 Check: What happens if $x = 4$ or $x = -3$ in $x^2 - x - 12$?

- This means 4 and 3 are roots of the polynomial $x^2 - x - 12$.
- The only way to get $x^2 - x - 12 = 0$ is for x to either be 4 or -3.
- We can rewrite

$$x^2 - x - 12 = (x - 4)(x + 3).$$

 Check: Verify the factorization on the right.

Quadratic Formula finds roots of 2nd degree polynomials



Quadratic Formula finds roots of 2nd degree polynomials

For any second degree polynomial

$$ax^2 + bx + c = 0,$$

the roots can be found using the **quadratic formula**:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

 Use the quadratic formula to find the solutions to

$$x^2 - x - 12 = 0.$$

Handwritten work for the quadratic formula applied to $x^2 - x - 12 = 0$:

Identifying coefficients: $a=1$, $b=-1$, $c=-12$.

$$-(-1) \pm \sqrt{(-1)^2 - 4(1)(-12)}$$

Calculation of the discriminant: $1 + 48$.

$$\frac{1 \pm \sqrt{49}}{2(1)}$$

Handwritten solutions:

1. $\frac{1 + 7}{2} = \frac{8}{2} = 4$

2. $\frac{1 - 7}{2} = \frac{-6}{2} = -3$

Applications

Population growth at discrete times

✏ A certain population size is measured at discrete time steps.

Notation:

N_0 = population at initial time

N_t = population after t time steps.

b = per capita birth rate

d = per capita death rate.

Assuming a closed population (no movement), we can model

growth by
$$N_{t+1} = N_t + bN_t - dN_t.$$



Population growth at discrete times

A certain population size is measured at discrete time steps. We use the notation:

- N_0 = population at initial time
- N_t = population after t time steps
- b = per capita birth rate
- d = per capita death rate

Assuming a closed population (no movement), we can model discrete growth by

$$N_{t+1} = N_t + bN_t - dN_t$$

subscripts
are
hard to see

 Solve these individually.

1. Simplify this expression by factoring out N_t and substituting $\lambda = 1 + b - d$.
2. Calculate N_1 , N_2 , and N_3 and express these in terms of N_0 . Find a pattern.
3. What is the general equation for the population size after T time steps? Your answer should only use λ , T , and N_0 .
4. What value of λ would give a stable population across time?
5. Find a value of λ for which the population grows and another for which it shrinks. More generally, for which values of λ does N shrink or grow?
6. Explain in your own words what is the role of λ in this model of population growth.

Population growth at discrete times


$$N_{t+1} = N_t + bN_t - dN_t = N_t (1 + b - d)$$

1. Simplify this expression by factoring out N_t , and substituting $\lambda = 1 + b - d$.
2. Calculate N_1, N_2 , and N_3 and express these in terms of N_0 . Find a pattern.
3. What is the general equation for the population size after T time steps? Your answer should only use λ , T , and N_0 .

1)
$$N_{t+1} = N_t (1 + b - d) = N_t \lambda = \cdot$$

$$N_{t+1} = \lambda N_t$$

2)
$$N_1 = N_{0+1} = \lambda N_0 \quad \Rightarrow \quad N_1 = \lambda N_0$$

$$N_2 = N_{1+1} = \lambda N_1 = \lambda (\lambda N_0) = \lambda^2 N_0$$

$$N_2 = \lambda N_1$$

$$\begin{aligned} N_1 &= \lambda N_0 \\ N_2 &= \lambda^2 N_0 \\ N_3 &= \lambda^3 N_0 \end{aligned}$$

Population growth at discrete times



Population after T time steps: $N_T = \lambda^T N_0$.

4. What value of λ would give a stable population across time?
5. Find a value of λ for which the population grows and another for which it shrinks. More generally, for which values of λ does N shrink or grow?
6. Explain in your own words what is the role of λ in this model of population growth.

$$\lambda = 1 + b - d \Rightarrow \text{if } \lambda = 1 \text{ then } 1 = 1 + b - d$$

$$\Rightarrow b = d.$$

$$\text{if } \lambda = 1, \text{ then } \underline{N_T} = 1^T \cdot N_0 = \underline{N_0}.$$

$$\begin{array}{l} \textcircled{5} \text{ growth } \lambda > 1 \\ \text{Shrink : } 1 > \lambda > 0. \end{array} \left. \vphantom{\begin{array}{l} \text{growth} \\ \text{shrink} \end{array}} \right\} \begin{array}{l} \lambda = 2 : N_T = 2^T N_0 \\ \lambda = \frac{1}{2} : N_T = \frac{1}{2}^T N_0 \end{array}$$

Population growth at discrete times

1) $N_{t+1} = N_t(1 + b - d) = \lambda N_t$

2) $N_1 = \lambda N_0, N_2 = \lambda^2 N_0, N_3 = \lambda^3 N_0$

3) $N_T = \lambda^T N_0$

4) $\lambda = 1$

5) $\lambda > 1$ grows (e.g. $\lambda = 1.1$); $0 \leq \lambda < 1$ shrinks (e.g. $\lambda = 0.9$)

6) Each time step multiplies N_t by λ , so λ is the finite (discrete-time) growth factor.

Wrapping up...

Get some extra practice!

Algebra problem set



What we covered today

-  **Algebra review**
-  **Solving equations**
-  **Exponents**
-  **Polynomials**
-  **Applications from MESM courses**